TasteBridge — Notebook Explainer

05_multi_seed_taste.ipynb • Multi-seed taste representation and candidate retrieval

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What this notebook is for

Every prior baseline notebook recommended tracks from a single seed track. Real listeners describe their taste with several songs at once, so this notebook is the first step toward that: it turns 3–5 liked tracks into one reusable **taste profile**, and uses that profile to pull a pool of candidate tracks worth recommending from. It deliberately stops before ranking those candidates into Familiar / Bridge / Adventurous tiers — that ranking step is scoped as the next notebook.

It reuses the exact embedding from the frozen Baseline 1 recipe (FMA Small → all 518 audio features → StandardScaler → PCA retaining 90% of variance → cosine distance) so that anything learned here is comparable to earlier results, and can later be re-run against a different feature space without changing the taste logic itself.

How a taste profile is built

Each seed track's feature vector is normalized to unit length (so comparisons are about direction/character, not loudness or magnitude). The profile keeps two things:

- **Centre** — the weighted average of the normalized seed vectors, re-normalized. This is a single convenient summary point, equivalent to "the middle of this listener's taste."
- **Individual seed vectors** — kept and used directly, not discarded after the centre is computed. The notebook is explicit about why: a listener who likes two distinct sounds (e.g. ambient techno and Jamaican dub) can have a centre that sits between them and resembles neither. Retaining each seed lets candidate retrieval anchor to what was actually liked, not just the average.

Candidate retrieval

For a given *per_anchor* setting (default tested at 150), the notebook runs a cosine nearest-neighbours search around the centre *and* around every individual seed, then takes the union of all of those results. Two similarity scores are recorded per candidate: similarity to the centre, and similarity to whichever seed it's closest to (*closest_seed_similarity*). This union approach is the mechanism that prevents a multi-genre taste from being flattened into its average.

Diagnostics before trusting the centre

Two checks confirm the centre isn't quietly hiding part of someone's taste:

- **Anchor counts** — how many retrieved candidates are closest to each seed. If one seed barely shows up, ranking by centre-similarity alone would under-serve that part of the listener's taste.
- **Seed-to-seed similarity** — a small matrix showing how similar the seeds are to each other. Low values flag that the listener's taste spans genuinely separate regions of the feature space.

Filtering before presenting results

Before any candidate is shown, the seed tracks themselves and every other track by a seed artist are excluded (case-insensitive, whitespace-trimmed name match). This stops the recommender from trivially "recommending" more songs by an artist the listener already named.

Worked example in the notebook

Three deliberately dissimilar seeds were used: **704** (Fósforo – *Mano De Dios*), **1083** (Magic Lantern – *Feasting on Energy*), and **56690** (Corty – *Trod Along*), each weighted equally.

Seed track	Candidates anchored to it	Share of 504
704 — Fósforo, Mano De Dios	160	31.7%
1083 — Magic Lantern, Feasting on Energy	154	30.6%
56690 — Corty, Trod Along	190	37.7%

504 distinct candidates came back before excluding seed artists. Retrieval was fairly even across the three seeds, with no seed badly underrepresented — the centre was not hiding a third of this listener's taste, though it did represent seed 1083 slightly less closely than the other two.

On manual listening to a small sample near each seed, most candidates sounded plausible together in a playlist. Some were straightforwardly familiar; others were less familiar but kept an audible connection to a seed — convincing early evidence for a **bridge track**, though the notebook notes these still need more differentiation before they'd read as a genuine step of discovery rather than a close cousin. Two tracks near seed 704 (Exteenager – *bonsoir tiroir!*, and The NSJ Crew – *Good Time*) were flagged as weak matches.

Caveat carried over from the notebook: this listening pass used the original 518-feature space (MFCC included, but no dedicated rhythm or percussive/harmonic-ratio features, since FMA's precomputed features.csv doesn't contain a tempo/beat family). Observations come from a small manual sample, not a scored evaluation across the whole candidate pool.

What the notebook explicitly leaves for next

Rank the artist-filtered candidate pool into Familiar, Bridge, and Adventurous lists, using centre similarity, closest-seed similarity, distance from the target, and list diversity together — not raw rank windows alone — then compare against the frozen single-track baseline on the same seed sets.

Generated as a plain-language walkthrough of notebooks/05_multi_seed_taste.ipynb for the TasteBridge capstone project.